

Idea Checker — MVP Product Requirements Document

Status: Draft for engineering handoff

1. Purpose & Vision

Idea Checker helps people validate whether a proposed solution actually solves a stated problem, using independent AI judgment instead of relying solely on the idea owner's own confidence. The long-term vision is a collaborative "GitHub for ideas" platform where solutions get stress-tested, refined, and validated by both AI and community consensus over time. The MVP exists to prove the narrow core of that value proposition: a person describes a problem, proposes a solution, and gets back a fast, structured, multi-perspective evaluation they can actually trust and act on.

2. Problem Statement

People constantly generate solutions to problems — business ideas, product features, process changes, technical approaches — but lack a fast, low-friction, objective way to sanity-check whether a given solution is actually good before committing time or money to it. Feedback from friends or colleagues tends to be biased or shallow. Formal validation (market research, expert consultation) is slow and expensive. A single AI model's opinion is also a known weak point, since individual models are inconsistent and prone to confident-sounding but wrong judgments. The gap is a tool that combines several independent AI judgments into one defensible score, fast enough to use at the moment an idea is formed.

3. Goals for MVP

- Let a person register, describe a problem, and propose one or more solutions to it.
- Produce a trustworthy evaluation of each solution by combining multiple independent AI judgments, never relying on a single model's output as the final answer.
- Score every solution consistently across five fixed dimensions: feasibility, effectiveness, scalability, cost efficiency, and innovation.
- Degrade gracefully — if some AI models fail or return unusable output, the evaluation should still complete using whatever valid responses remain, rather than failing outright.
- Give the user a result they can act on: a numeric score plus enough specific written explanation to understand *why*, not just *what*.
- Keep each user's problems, solutions, and evaluation history private and tied to their account.

4. Non-Goals for MVP (explicitly deferred)

- Scenario-based "pressure testing" or stress simulations (e.g., "what if the budget is cut 50%").
- Community features: voting, commenting, forking, leaderboards, or any visibility of one user's content to another.
- Editing or deleting a problem or solution after creation (MVP is create/read only).
- Re-evaluation workflows or evaluation history — a solution gets exactly one evaluation in MVP.
- Dashboards, analytics, search, filtering, or recommendations.
- Any "memory" feature that references a user's past ideas, or relationships between different problems.
- Non-English input/output, localization.
- Native mobile apps.

5. Target Users

- **Solo founder / indie hacker** — has more ideas than time and wants a fast gut-check before committing to one.
- **Student / hackathon participant** — needs to defend a solution's feasibility quickly, often under time pressure.
- **Product manager / consultant** — wants a structured, defensible second opinion to bring into a stakeholder conversation.

Across all three, the shared need is speed plus a credible sense of objectivity — a score that feels like more than one person's opinion.

6. Core User Flow

1. User creates an account or logs in.
2. User creates a Problem: title, description, optional tags.
3. User submits one or more Solutions against that Problem: free-text description of the proposed approach.
4. User requests an Evaluation for a specific Solution.
5. The system gathers judgments from multiple independent AI evaluators in parallel, scores the solution across the five dimensions, aggregates the results, and persists the evaluation.
6. User views the result: five sub-scores, one overall score, and written feedback explaining the reasoning.

7. User may submit another solution to the same problem (or a new problem) and repeat from step 3 to compare results.

7. Functional Requirements

7.1 Accounts & Identity

- **FR-1:** A new user can register with, at minimum, an email and password.
- **FR-2:** A returning user can log in and stay authenticated across a session without re-entering credentials for every action.
- **FR-3:** Passwords are never stored or transmitted in plain text.
- **FR-4:** Every Problem and Solution is owned by exactly one user — its creator.
- **FR-5:** A user can only view or act on Problems and Solutions they own (see open question in §13).
- **FR-6:** Invalid registration input (malformed email, empty password, etc.) is rejected with a clear, specific error rather than a generic failure.

7.2 Problems

- **FR-7:** An authenticated user can create a Problem with a title, a description, and optional free-text tags.
- **FR-8:** Title and description must meet a minimum length before being accepted — a one-word "problem" should be rejected, since it produces meaningless evaluations downstream.
- **FR-9:** A user can list all Problems they own.
- **FR-10:** A user can retrieve a single Problem they own by its identifier.

7.3 Solutions

- **FR-11:** An authenticated user can submit a Solution against a Problem they own, with free-text content describing the proposed approach.
- **FR-12:** Solution content must meet a minimum length before acceptance, for the same reason as FR-8.
- **FR-13:** A user can list all Solutions submitted against a given Problem.
- **FR-14:** A Solution always belongs to exactly one Problem; it cannot exist without a parent Problem.

7.4 AI Evaluation (core differentiator)

- **FR-15:** A user can request an Evaluation for any Solution they own.

- **FR-16:** The system must independently query more than one AI model with the Problem and Solution as context. The final result must never rely on a single model's response alone.
- **FR-17:** Each responding model's output is scored against five fixed dimensions — feasibility, effectiveness, scalability, cost efficiency, and innovation — each on a 0–10 scale.
- **FR-18:** If a model's response can't be parsed into valid scores, or the model fails to respond at all, that model is excluded from the aggregate. It must never cause the whole evaluation to fail.
- **FR-19:** The final score for each dimension is the average of all valid (successfully parsed) scores for that dimension across responding models.
- **FR-20:** The overall solution score is derived from the five dimension scores and presented on a 0–100 scale.
- **FR-21:** If every model fails to produce a valid response, the system returns a clear, user-facing error — never a fabricated or zeroed-out score.
- **FR-22:** Alongside the numeric scores, the evaluation includes a short, specific written explanation of the solution's strengths and weaknesses. This is core to the product's value, not optional polish — a static or generic message does not satisfy this requirement.
- **FR-23:** Each Evaluation is permanently linked to the Solution it ran against and is retrievable later.
- **FR-24:** A Solution has at most one Evaluation in MVP — no re-evaluation history.

7.5 Viewing Results

- **FR-25:** A user can retrieve the Evaluation for a Solution they own, including all five sub-scores, the overall score, and the written feedback.
- **FR-26:** If a Solution hasn't been evaluated yet, the system communicates that clearly rather than returning an empty or malformed result.

8. Non-Functional Requirements

- **NFR-1 (Resilience):** The evaluation pipeline tolerates individual AI model timeouts, errors, or malformed output without failing the entire request (ties to FR-18).
- **NFR-2 (Latency):** A full evaluation request, including all parallel model calls, should complete within a time the user will actually wait for — target well under 15 seconds under normal conditions, not minutes.

- **NFR-3 (Data isolation):** One user can never read, modify, or evaluate another user's Problems or Solutions.
- **NFR-4 (Input validation):** All user-supplied content is validated for length and required fields before being accepted or sent to any AI model, to avoid wasting AI calls on garbage input.
- **NFR-5 (Abuse / cost control):** The system guards against a single user triggering an unreasonable volume of evaluation requests in a short window, since every evaluation consumes paid third-party AI calls.
- **NFR-6 (Error transparency):** Failures — a model down, a malformed response, an invalid request — surface as clear, specific messages, never a raw internal error or silent failure.
- **NFR-7 (Auditability):** The system retains enough metadata about each evaluation (e.g., how many models succeeded vs. failed) to let the team diagnose quality or reliability issues later, even if this isn't shown to end users in MVP.
- **NFR-8 (Forward compatibility):** None of the above should block the later phases in §4 — e.g., the data model should be able to support a Solution eventually having multiple Evaluations, even though MVP only uses one.

9. Data Model (conceptual)

- **User** — identifier, name, email (unique), credential, role, account creation timestamp.
- **Problem** — identifier, title, description, tags, owning user, creation timestamp.
- **Solution** — identifier, content, owning user, parent problem (required), creation timestamp.
- **Evaluation** — identifier, parent solution (one-to-one in MVP), five dimension scores, overall score, written feedback, creation timestamp, and a record of how many models contributed a valid response (supports NFR-7).

Relationships: one User has many Problems; one Problem has many Solutions; one Solution has at most one Evaluation in MVP.

10. Evaluation Logic — Detail

- Each of the five dimensions is independently scored 0–10 by each responding model.
- A score is "valid" if it's a well-formed number. A number outside 0–10 (e.g., a model returning 12) is clamped to the nearest valid bound rather than discarded — only a missing or non-numeric response is treated as invalid for that dimension. Discarding

a whole response over a minor formatting slip wastes a model call that's otherwise usable.

- The aggregated score per dimension is the arithmetic mean of all valid scores for that dimension across responding models.
- The overall score is the mean of the five aggregated dimension scores, scaled to a 0–100 display range.
- Minimum viable consensus: at least one model must return a valid response for the evaluation to succeed (FR-21). Whether a higher bar (e.g., at least 2 of 3) is needed for a result to count as "trustworthy" is an open product decision — see §13.

11. Success Metrics for MVP

- **Evaluation completion rate** — percentage of evaluation requests that return a valid result despite individual model failures (directly measures the resilience goal).
- **Time-to-result** — median and 95th-percentile time from request to result.
- **Activation rate** — percentage of registered users who create a Problem, submit a Solution, and request an evaluation within their first session.
- **Repeat usage** — percentage of users who submit a second Solution after seeing their first evaluation, as a proxy for whether the result felt useful enough to return.
- **Perceived usefulness** — qualitative tracking (support requests, direct feedback) of whether the written feedback (FR-22) reads as specific and trustworthy rather than generic.

12. Assumptions & Constraints

- Problems and Solutions are private to their creator in MVP; nothing is visible to other users until the community phase.
- Input and output are assumed to be in English.
- Users are assumed to have connectivity sufficient to wait several seconds for an evaluation to complete.
- The product depends on third-party AI providers being reachable; a sustained outage across all of them is not something MVP is expected to fully solve beyond the minimum failure behavior in FR-21.
- "Problem" and "Solution" are generic free text — MVP does not restrict or validate by domain (business, technical, policy, etc.).

13. Open Questions

- Should account creation be required before trying the evaluation flow, or should a limited guest evaluation be allowed to reduce signup friction?
- Should the five dimensions be weighted equally in the overall score, or should some (e.g., feasibility) count more?
- Is one successful model response enough to call a result "trustworthy" (current floor), or should the bar be higher?
- Should there be a per-user or global rate limit on evaluation requests, given each one has a real cost?
- Should owners be able to soft-delete a Problem or Solution in MVP, or is that fully deferred as currently scoped?

14. Definition of Done for MVP

- A user can register, log in, and stay authenticated across a session.
- A user can create a Problem and submit at least one Solution against it.
- A user can request an Evaluation and receive five dimension scores, an overall 0–100 score, and specific written feedback — even when one or more underlying AI models fail.
- A user can retrieve a past Evaluation for any Solution they own.
- No user can view or act on another user's Problems, Solutions, or Evaluations.
- Every requirement in §7–8 is met, and every open question in §13 has been explicitly resolved rather than left ambiguous in the shipped product.